

AE 20: Assessing Logistic Regression Models

! Important

- Open [RStudio](#) and create a subfolder in your AE folder called “AE-20”.
- Go to the [Canvas](#) and locate your AE-20 assignment to get started.
- Upload the `ae-20.qmd` and `framingham.csv` files into the folder you just created. The `.qmd` and PDF responses are due in Canvas. You can check the due date on the Canvas assignment.

Packages

```
library(tidyverse)
library(broom)
library(ggformula)
library(mosaic)
library(knitr)

heart_disease <- read_csv("framingham.csv") |>
  select(totChol, TenYearCHD, age, BMI, cigsPerDay, heartRate, sysBP, diabetes) |>
  drop_na()
```

Data: Framingham study

This data set is from an ongoing cardiovascular study on residents of the town of Framingham, Massachusetts. We want to predict if a randomly selected adult is high risk for heart disease in the next 10 years.

Response variable

- TenYearCHD:
 - 1: Patient developed heart disease within 10 years of exam
 - 0: Patient did not develop heart disease within 10 years of exam

What's my predictor variable?

Based on your group, use the following as your predictor variable.

- Group 1 - totChol: total cholesterol (mg/dL)
- Group 2 -BMI: patient's body mass index
- Group 3 -cigsPerDay: number of cigarettes patient smokes per day
- Group 4 -heartRate: Heart rate (beats per minute)

Additional Variables

- sysbp - the patients systolic blood pressure at the time of examination
- diabetes - 1 if the patient had diabetes and 0 if the patient didn't have diabetes at the time of examination

Exercise 0

Fit a logistic regression model predicting TenYearCHD from your group's predictor variable. Note that we won't return to this model until Exercise 3.

Exercise 1

Before you work on your own model, let's consider the variable sysBP which represents the patients systolic blood pressure (the top number). You have 120 seconds, find the β 's that result in the largest log-likelihood for a logistic regression model predicting the risk of coronary heart disease from sysBP:

```
# Change these
beta0 <- 1
beta1 <- 1

# Don't change anything below this line
```

```

predicted_probabilities <- exp(beta0 + beta1*heart_disease$sysBP)/(1+exp(beta0 + beta1*heart.
log_likelihoods <- heart_disease$TenYearCHD*log(predicted_probabilities) +
                    (1-heart_disease$TenYearCHD)*log(1-predicted_probabilities)

# Final log likelihood
sum(log_likelihoods)

```

[1] NaN

Exercise 2

Compute the empirical logit for each level of **diabetes**:

1. Use the function **tally** to compute the count the number of successes and failures for each level of **diabetes**.
2. Compute the empirical odds.
3. Compute the log of these odds.

Exercise 3

Is linearity satisfied for the model you fit in Exercise 0?

Exercise 4 (Time Permitting)

On a whiteboard, design a simulation-based hypothesis test for determining whether the coefficient of your explanatory variable is a statistically significant predictor for the risk of **TenYearCHD**.

Submission

! Important

To submit the AE:

- Render the document to produce the PDF with all of your work from today's class.
- Upload your QMD and PDF files to the Canvas assignment.